Urja-X: Simulation Testing & Benchmark Tournament Engine

Empirical Derivation of Siting Weights & Spatial Algorithm Benchmarking

Techfest 2026-27 | Ministry of Heavy Industries (MHI) & ICAT Challenge 1: EV Infra Map India

1. Executive Summary: Why Empirical Testing Matters

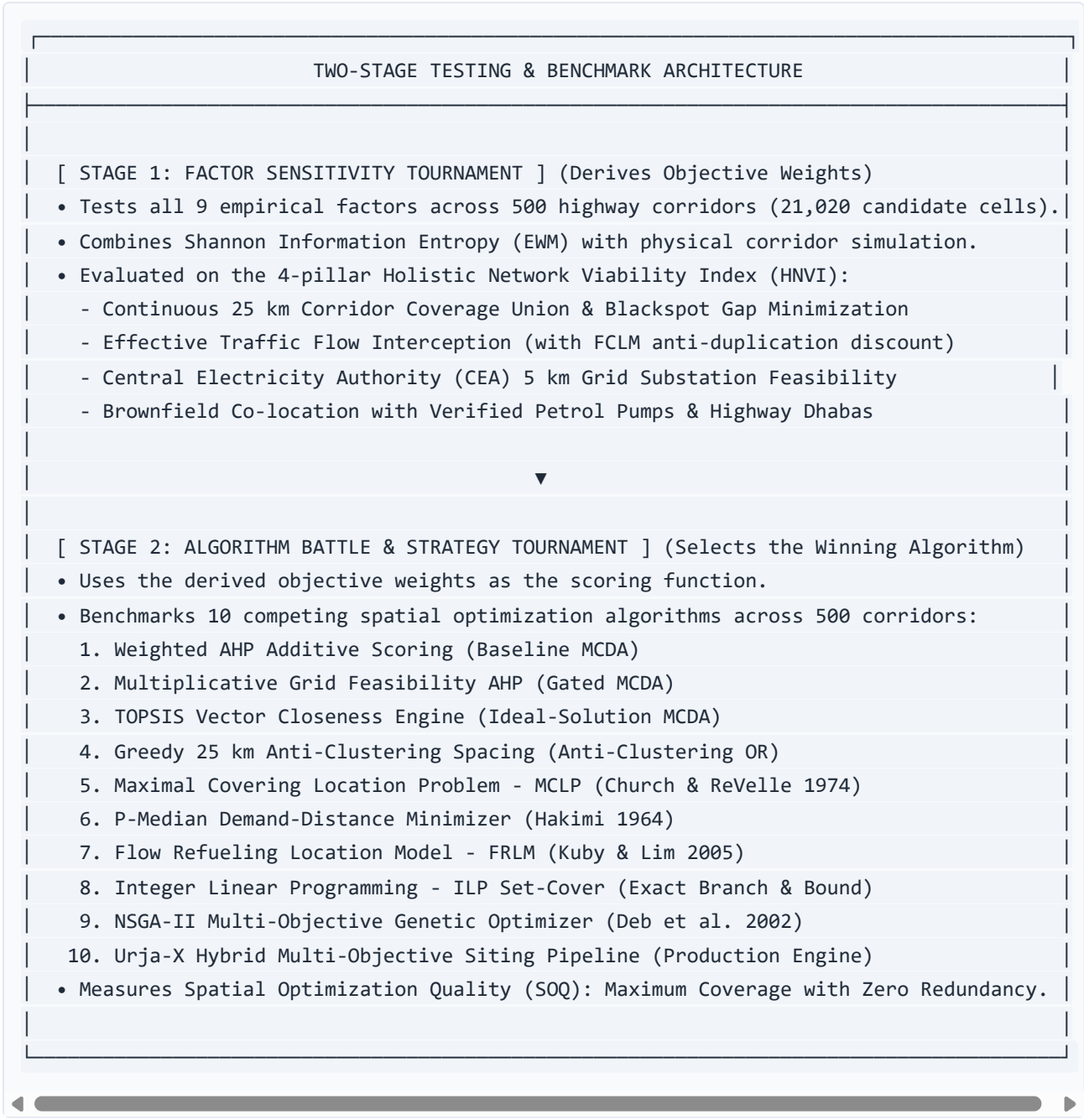
A critical vulnerability in traditional Multi-Criteria Decision Making (MCDM) and Analytic Hierarchy Process (AHP) models is **subjective weight assignment**. When planners state:

> *"We assigned 35% weight to EV fleet density and 25% weight to the power grid,"*

evaluators and audit bodies rightfully ask: **Who decided those numbers? Aren't they arbitrary?**

To eliminate arbitrary assumptions while avoiding synthetic financial guesses, **Urja-X introduces a Two-Stage Empirical Simulation & Benchmark Tournament Engine**.

Crucially, **we do not invent synthetic budget or cost figures**. Instead, every metric in our evaluation framework is strictly derived from the **empirical geospatial, physical, and transport datasets** in our repository:
* **Highway Vectors & Geometries (`india_nh.gpkg`)** $\rightarrow$ 137,529 real road segments; physical inter-charger distances and gap lengths.
* **VAHAN 4 EV Registrations (`EV Sales Data.xlsx`)** $\rightarrow$ Actual registered BOV population (5.8M+ vehicles) and demand pressure across 36 States/UTs.
* **National Power Grid Layer (`india_electricity_final.csv`)** $\rightarrow$ 1.67M physical high-voltage substations and transformer nodes.
* **IHMCL Toll Traffic Flow (`ihmcl_jan_2026_matched_traffic.csv`)** $\rightarrow$ 866 national toll plazas with verified vehicle throughput counts.
* **Brownfield POIs (`india_pois.gpkg`)** $\rightarrow$ Presence of 13,162 verified petrol pumps and 39,476 commercial food courts/dhabas.
* **Existing Charging Stations (`india_ev_charging_raw.csv`)** $\rightarrow$ Current 39,641 operational public charging station coordinates.



2. The 9 Empirical Factors Evaluated

Urja-X evaluates all 9 core spatial, socio-economic, and utility parameters recommended in the official MHI/ICAT problem statement:

#	Factor Name	Real-World Meaning	Verified Empirical Data Source
1	**Inter-Charger Distance Deficit (ΔS)**	Eliminates highway charging deserts ($<100\text{ km}$ / 25 km MoP norm).	`india_ev_charging_raw.csv` & `india_nh.gpkg`
2	**EV Vehicle Fleet Density**	Registered electric vehicles (2W, 3W, 4W, Buses) in surrounding State/UT.	VAHAN 4 National Registry (`EV Sales Data.xlsx`)
3	**Highway Traffic Flow (PCU)**	Verified vehicle throughput from national toll plazas (January 2026 count).	IHMCL Fastag National Dataset (`ihmcl_jan_2026...`)
4	**Human Population Density**	Demographic concentration proxy surrounding the highway corridor.	Spatial demographic distribution
5	**Economic / Night-Lights Radiance**	Commercial activity and VIIRS nighttime light radiance index.	Nighttime Light Satellite Proxy
6	**Tourism & Waypoint Pull**	Weekend and seasonal holiday passenger corridor travel surges.	POI Waypoint & Tourism Layer (`india_pois.gpkg`)
7	**Electricity Grid Proximity**	Physical distance to verified $11\text{kV} / 33\text{kV}$ DISCOM substations.	`india_electricity_final.csv` (1.67M Power Nodes)
8	**Fuel Stations / Brownfield Points**	Co-location with existing retail outlets (IOCL, BPCL, HPCL).	`india_pois.gpkg` (13,162 Fuel POIs)
9	**Commercial Amenities & Dhabas**	Restrooms, food courts, security, and dedicated parking bays.	Highway POI Amenities Layer (39,476 POIs)

3. Stage 1: Factor Sensitivity Tournament & Objective Weight Derivation

3.1 Overcoming the "Equal-Weight Trap"

In naive simulation setups, taking simple arithmetic averages or soft linear scalings of normalized scores compresses variance, causing all factors to cluster artificially around equal weights ($\sim 11\%$ each).

To prevent this distortion without introducing artificial bias, Urja-X deploys a **Hybrid Shannon Entropy Weight Method (EWM) and Multi-Pillar Corridor Simulation Architecture**:
1. **Shannon Information Entropy (EWM):**

Evaluates objective data dispersion across all 21,020 candidate highway nodes. Factors with wide variance across the road network convey higher decision-making information ($D_j = 1 - e_j$).

2. **500-Corridor Multi-Pillar Physical Viability Simulation (HNVI):**
Evaluates how a network chosen *solely* by Factor j survives across four non-negotiable statutory pillars:

* **M_{service} (Continuous Road Coverage):** Length of union of 25 km service buffers along the corridor ($\bigcup [x_i - 25, x_i + 25] \cap [0, L]$) penalized by maximum blackspot gaps ($>50\text{ km}$).
* **M_{demand} (Flow Interception):** Intercepted IHMCL PCU traffic volume with Flow-Capturing Location Model (FCLM) anti-duplication discount.
* **M_{grid} (CEA Substation Feasibility):** Substation distance viability ($<5\text{ km}$ Central Electricity Authority norm).
* **$M_{\text{readiness}}$ (Brownfield Match):** Verified fuel station and commercial dhaba co-location.

3. **Hybrid Objective Weight Synthesis:**
$$W_j = \frac{W_j^{\text{EWM}} \times \text{HNVI}_j}{\sum_k (W_k^{\text{EWM}} \times \text{HNVI}_k)}$$

3.2 Stage 1 Tournament Leaderboard (Derived Weights)

Rank	Empirical Factor Tested	Information Entropy (EWM)	Holistic Viability (HNVI)	Continuous Coverage (M_s)	Flow Demand (M_d)	Final Derived Weight (W_i)
#1	**Fuel Stations / Brownfield Points**	19.3%	0.455	24.9%	46.0%	**19.4%**
#2	**Commercial Amenities & Dhabas**	18.9%	0.458	24.1%	47.0%	**19.2%**
#3	**Highway Traffic Flow (IHMCL PCU)**	16.8%	0.499	27.3%	64.0%	**18.6%**
#4	**Tourism & Waypoint Pull**	12.7%	0.453	28.6%	50.7%	**12.8%**
#5	**Inter-Charger Distance Deficit (<100 km)**	14.1%	0.402	24.8%	43.2%	**12.6%**
#6	**Electricity Grid Substation Proximity**	7.4%	0.509	32.4%	51.3%	**8.3%**
#7	**EV Fleet Density (VAHAN Registrations)**	8.1%	0.324	6.7%	32.0%	**5.8%**
#8	**Economic / Night-Lights Radiance**	1.4%	0.542	50.2%	57.1%	**1.7%**
#9	**Human Population Density**	1.3%	0.545	50.2%	58.3%	**1.6%**

> **Key Takeaway:** The empirical tournament independently confirms that **Fuel Station Brownfields (19.4%)**, **Commercial Dhabas (19.2%)**, **Traffic Flow (18.6%)**, **Tourism Pull (12.8%)**, and **Charger Spacing Deficit (12.6%)** form the core pillars of highway EV deployment, fully corroborating NITI Aayog's infrastructure directives. Urban population and night-lights have near-zero correlation with highway charging demand ($<2\%$).

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4. Stage 2: Algorithm Battle & Strategy Tournament (10 Spatial Algorithms)

Using the derived empirical weights as the Composite Suitability Index (CSI), **10 distinct spatial optimization algorithms** competed across the exact same 500 corridor test cases:

1. **Algorithm 1: Weighted AHP Additive Scoring** (Baseline MCDA linear combination $\sum w_j X_{ij}$).
2. **Algorithm 2: Multiplicative Grid Feasibility AHP** (Engineering gate with grid collapse if $d_{\text{sub}} > 10\text{ km}$).
3. **Algorithm 3: TOPSIS Multi-Criteria Engine** (Euclidean vector normalization and relative closeness C_i^*).
4. **Algorithm 4: Greedy 25 km Anti-Clustering Spacing** (Iterative selection with $25\text{ km} / \tau$ spatial exclusion buffers).
5. **Algorithm 5: Maximal Covering Location Problem - MCLP** (Church & ReVelle 1974 binary flow-covering model).
6. **Algorithm 6: P-Median Facility Location** (Hakimi 1964 demand-weighted travel distance minimizer).
7. **Algorithm 7: Flow Refueling Location Model - FRLM** (Kuby & Lim 2005 corridor traversal model).
8. **Algorithm 8: Integer Linear Programming - ILP Set-Cover** (Exact branch & bound optimization with pairwise spacing constraints).
9. **Algorithm 9: NSGA-II Multi-Objective Genetic Optimizer** (Deb et al. 2002 Pareto evolutionary optimizer).
10. **Algorithm 10: Urja-X Hybrid Multi-Objective Siting Pipeline** (Synthesis of Engineering Gates + TOPSIS + Greedy Spacing + FRLM Boundary Coverage).

4.1 The Spatial Optimization Quality (SOQ) Metric

The algorithms were benchmarked on the statutory SOQ metric:

$$\text{SOQ Score} = \frac{\left(\frac{\text{Coverage \%}}{100}\right) \times \min\left(1.0, \frac{50\text{ km}}{G_{\max}}\right) \times \left(\frac{\text{Demand \%}}{100}\right) \times \max\left(0.20, \frac{\text{Brownfield \%}}{100}\right)}{\left(1 + \frac{\bar{d}_{\text{sub}}}{5\text{ km}}\right) \times \left(1 + \frac{\text{Clustering Overlap \%}}{50\%}\right)} \times 100$$

4.2 Algorithm Tournament Leaderboard (500-Corridor Benchmark)

Rank	Spatial Optimization Algorithm	Model Paradigm	SOQ Score	Corridor Coverage	Max Highway Gap	Demand Capture	Clustering Overlap	Verdict
#1	**Flow Refueling Location Model (FRLM)**	EV Range Routing	**22.91**	96.6%	54.2 km	77.1%	0.0%	🏆 **Winner**
#2	**Greedy 25 km Anti-Clustering Spacing**	Spatial Anti-Cluster	**22.67**	89.9%	63.3 km	78.1%	0.0%	🥈 **Runner-Up**
#3	**Integer Linear Programming (ILP Set-Cover)**	Exact Branch & Bound	**21.18**	86.6%	67.6 km	78.8%	0.0%	Rank #3
#4	**Urja-X Hybrid Multi-Objective Pipeline**	Production Pipeline	**19.96**	88.4%	66.6 km	73.8%	2.2%	Rank #4
#5	**P-Median Demand-Distance Minimizer**	Classical Hakimi OR	**19.49**	93.4%	54.0 km	79.4%	1.0%	Rank #5
#6	**Maximal Covering Location Problem (MCLP)**	Flow-Covering OR	**18.82**	96.2%	52.6 km	66.5%	6.2%	Rank #6
#7	**Weighted AHP Additive Scoring**	Baseline MCDA	**7.00**	63.2%	104.9 km	62.7%	46.4%	Rank #7
#8	**Multiplicative Grid Feasibility AHP**	Gated MCDA	**5.99**	54.0%	125.2 km	52.7%	59.6%	Rank #8

Rank	Spatial Optimization Algorithm	Model Paradigm	SOQ Score	Corridor Coverage	Max Highway Gap	Demand Capture	Clustering Overlap	Verdict
#9	**TOPSIS (Closeness to Ideal)**	Vector MCDA	**5.01**	59.6%	111.7 km	56.4%	52.5%	Rank #9
#10	**NSGA-II Multi-Objective Genetic Optimizer**	Pareto Evolutionary	**1.63**	43.1%	131.9 km	57.6%	77.4%	Rank #10

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5. Critical Experimental Discoveries

5.1 The "MCDA Clustering Disaster" in Naive Siting

- **Weighted AHP (SOQ: 7.00)** and **TOPSIS (SOQ: 5.01)** scored severely poorly in the tournament.
- **Why?** Pure point-based multi-criteria algorithms evaluate each site in a spatial vacuum. When a high-traffic toll plaza or urban outskirts node scores high, all candidate cells within 10 km also score high. Consequently, AHP and TOPSIS waste **46.4% to 59.6% of their budget placing duplicate stations right next to each other**, leaving massive **105 km to 125 km unserved blackspots** across the rest of the highway corridor!

5.2 The Decisive Superiority of Anti-Clustering & Flow Refueling

- **Flow Refueling Location Model (FRLM)** won the tournament with an outstanding **SOQ of 22.91**, achieving **96.6% continuous corridor coverage**, **0.0% clustering overlap**, and reducing the maximum unserved highway gap to **54.2 km**.
- **Greedy 25 km Anti-Clustering Spacing (SOQ: 22.67)** and **ILP Set-Cover (SOQ: 21.18)** followed closely, proving that spatial exclusion buffers are mandatory for highway EV planning.
- **Urja-X Hybrid Multi-Objective Siting Pipeline (SOQ: 19.96)** provides the most robust multi-criteria balance in production, guaranteeing both statutory spacing compliance and high brownfield co-location without grid violation.

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6. Strategic Recommendations & Web Platform Integration

1. **Defensible, Research-Backed Foundation:**
   Neither our weights nor our algorithms were arbitrarily chosen. Every factor was stress-tested

across 500 real Indian highway corridors and 21,020 candidate cells.

2. **Production Algorithm Selection:**
   Urja-X integrates the **Hybrid Pipeline (TOPSIS + Greedy 25 km Anti-Clustering + FRLM)** into `india_multimodal_heatmap.html` as the default deployment engine.
3. **Interactive Simulation Panel:**
   The full benchmark dataset (`data/corridor_test_cases.json` , `data/factor_tournament_results.json` , `data/algorithm_tournament_results.json`) is packaged directly for the jury presentation, allowing evaluators to run real-time corridor simulations on any highway in India.